GASEOUS STATE

1. $PV = nRT$ $\qquad PM = dRT$
   ↳ free space available

2. Boyle → T const. $\qquad$ Charle's → P const. $\qquad$ Gay → V const. $\qquad$ Avogadro → T & P const.

3. 1 atm = 1 torr = 76 cm of Hg = 1.013 bar = 1.013×10^5 N/m²

4. $R = 0.0821$ atm L $mol^{-1} K^{-1}$
   $= 8.314$ J $mol^{-1} K^{-1}$
   ≈ 2 cal $mol^{-1} K^{-1}$ $\qquad (\approx 1.98)$

5. Graham's Law → $r \propto \frac{1}{\sqrt{d}}$ (r → rate of diffusion, d → density of gas)

   → ① $r \propto A$ $\qquad$ ② $r \propto P_{gas}$ $\qquad$ ③ $r \propto \frac{1}{\sqrt{T}}$ $\qquad$ ④ $r \propto \frac{1}{\sqrt{M}}$

6. KTG → ⓐ $PV = \frac{1}{3} mN (V_{RMS})^2$

   ⓑ → $V_{RMS} = \sqrt{\frac{3RT}{M}}$ $\qquad \rightarrow V_{MPS} = \sqrt{\frac{2RT}{M}}$ $\qquad \rightarrow V_{avg} = \sqrt{\frac{8RT}{\pi M}}$

   ⓒ Total KE $= \frac{3}{2} PV = \frac{3}{2} nRT$

   ⓓ Avg KE per molecule $= \frac{3}{2} kT$

7. $k = \frac{R}{N_A}$

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8. Real Gas → $\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$

(a: nature of $IMF_{attrac.}$; b: size of particles)

$b = 4 \times N_A \times \frac{4}{3}\pi r^3$

9. $z = \frac{V_{(I\ want)}}{V_{(I\ have)}}$ → ideal gas ka vol.

a) P↓ ⇒ I want to reduce my vol. (attrac. forces)
↳ $V_{(I\ want)} < V_{(I\ have)}$ ∴ $z < 1$

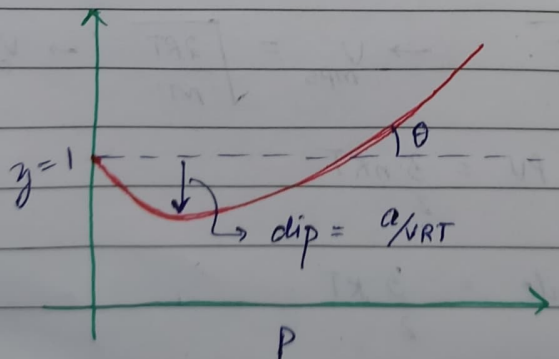
$z = \frac{PV_m}{RT}$

b) P↑ ⇒ Mujhe failna hai (Repulsive forces)
↳ $z > 1$

c) P↓↓↓ ⇒ Mere molecules itne duur duur hain ki na repulsive force hai na attract.
$z = 1$. I am ideal!

10. A real gas behaves ideal at
↳ P↓↓↓ → a→0 and V→∞ so V-nb→V
↳ P↓ + T↑
V-nb≃V ← V↑ ↵ ↳ mol move so rapidly, a→0

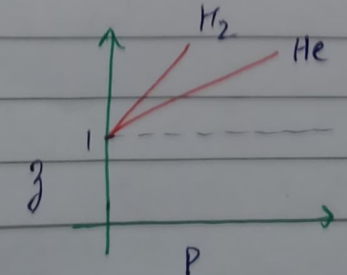
11.

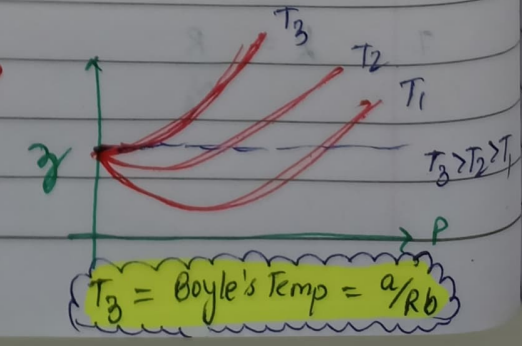


$\tan\theta = \frac{b}{RT}$

a↑ ⇒ IMF↑ ⇒ z mein dip↑

12. H_2 & He





T_3 = Boyle's Temp = a/Rb

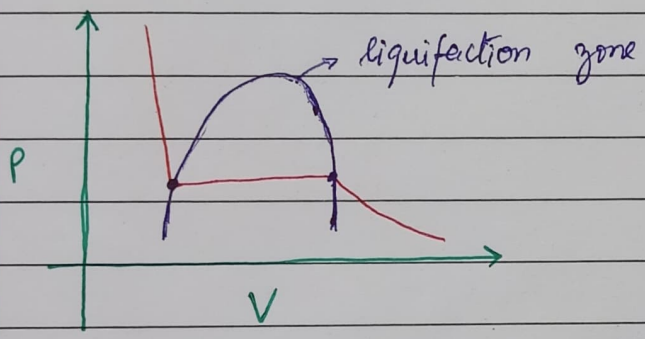
13. Gas → above T_c
Vapour → below T_c
(which can be liquified)

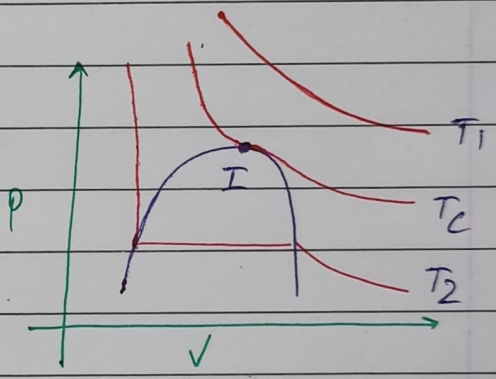
14. T_c → critical temp. → Temp above which a gas xxx be liquified.

P_c → Min Pr. required to liquify a gas at T_c

V_c → Vol. at T_c and P_c.

15.





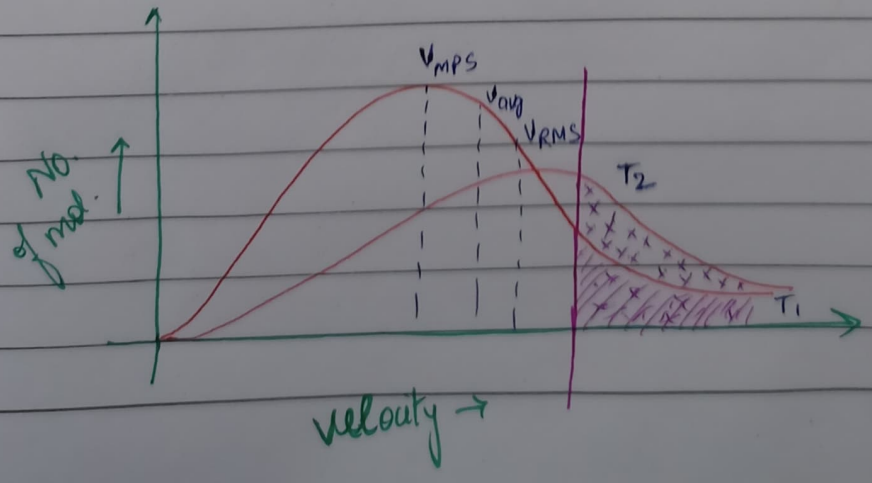
$T_1 > T_c > T_2$

I → point of inflection

$\frac{\partial^2 P}{\partial V^2}$ and $\frac{\partial P}{\partial V} = 0$

16. $T_c = \frac{8a}{27Rb}$ $P_c = \frac{a}{27b^2}$ $V_c = 3b$

17. Maxwell's Distr. Curve



$T_2 > T_1$
↳ more energy
↳ more (higher) velocities